

CAPS ANSWER KEY

ANSWER KEY (CAPS-1)

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|---------|---------|---------------------------------------|--|---------|
| 1. (A) | 2. (B) | 3. (B) | 4. (C) | 5. (A) |
| 6. (BD) | 7. (BC) | 8. (ACD) | 9. (B) | 10. (D) |
| 11. (B) | 12. 9 | 13. 1 | 14. 4 Sec | 15. 7 |
| 16. 4 | 17. 3 | 18. $v' = \sqrt{\frac{u^2 + v^2}{2}}$ | 19. $CD = \frac{\ell}{\sqrt{n^2 - 1}}$ | |

20. (a) $v = \alpha^2 t / 2$, $a = \alpha^2 / 2$; (b) $\langle v \rangle = \frac{\alpha \sqrt{s}}{2}$ 21. (A) -p, (B) -v, (C) -s, (D), -q, (E) -u, (F) -t
22. (A) P, T (B) R (C) P, Q (D) Q, R, S 23. 9 min, 40 km hr⁻¹
24. 7/2 s 25. $2mv_0^{1/2} = d$

ANSWER KEY (CAPS-2)

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|-----------|------------|----------------------|---------|---------|
| 1. (C) | 2. (B) | 3. (AB) | 4. (AC) | 5. (AD) |
| 6. (BCD) | 7. (BCD) | 8. (AC) | 9. (AC) | 10. (D) |
| 11. (C) | 12. (A) | 13. (C) | 14. (B) | 15. (D) |
| 16. 3 | 17. (0002) | 18. 5 | 19. 4 | |
| 20. 10sec | 21. 3 | 22. 3 km/h towards B | 23. 8 | |
| 24. 5 | 25. 4 m/s | | | |

ANSWER KEY (CAPS-3)

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|---|--|--|-----------|----------|
| 1. (A) | 2. (C) | 3. (A) | 4. (A) | 5. (A) |
| 6. (B) | 7. (C) | 8. (A) | 9. (D) | 10. (B) |
| 11. (A) | 12. (A) | 13. (ABCD) | 14. (ABC) | 15. (AD) |
| 16. (ABC) | 17. (A) | 18. (B) | 19. (A) | |
| 20. $t = 2 \text{ cm}$ | 21. (50.00) | 22. (A) - q,t; (B) - p,r,s ; (C) - p,r,s ; (D) - p,r,t | | |
| 23. $\left(\frac{135}{6}\right) \text{ cm} = 22.5 \text{ cm}$ below the upper surface of the glass slab | | | | |
| 24. (i) No shift is observed (ii) 1 cm | 25. $y = t\theta \left[1 - \frac{1}{n}\right]$. | | | |

ANSWER KEY (CAPS-4)

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|---|---|-----------------------------------|-----------------------------|----------|
| 1. (B) | 2. (B) | 3. (C) | 4. (D) | 5. (B) |
| 6. (D) | 7. (C) | 8. (C) | 9. (D) | 10. (C) |
| 11. (AC) | 12. (ABCD) | 13. (ABC) | 14. (AC) | 15. (D) |
| 16. (B) | 17. (D) | 18. (0002) | 19. $\sqrt{7}u \text{ m/s}$ | 20. 30 N |
| 21. (A) - p,s ; (B) - p,t ; (C) - p,t ; (D) - p,r | 22. $a = \frac{(m - m')g}{2M + 3m + 3m'}$ | | | |
| 23. (a) $v_{\max} = \sqrt{\frac{g \sin \theta \tan \theta}{\mu_0}}$ (b) $x = \frac{\tan \theta}{\mu_0}$ | 24. $F_{fr} = 2mM\ell / (M - m)t^2$ | 25. $\frac{u^2}{b} \tan^3 \theta$ | | |

ANSWER KEY (CAPS-5)

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|------------|-------------------------------|---------|-----------|---------|
| 1. (A) | 2. (B) | 3. (D) | 4. (A) | 5. (A) |
| 6. (B) | 7. (C) | 8. (B) | 9. (B) | 10. (C) |
| 11. (BC) | 12. (ABC) | 13. (B) | 14. (C) | 15. (B) |
| 16. (0004) | 17. $\frac{5}{8} \text{ mga}$ | 18. 2 | 19. n^3 | |

20. (A) T (B) P (C) Q, R (D) Q 21. $x = \sqrt{\frac{(3M+m)}{3k}} v.$

22. $F = \frac{5}{12} mg$ 23. $\frac{b^2}{4a}$ 24. $\frac{3mg}{2k}$ 25. $\left[m_1 + \frac{m_2}{2} \right] \mu g.$

ANSWER KEY (CAPS-6)

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|-------------------------------------|---------------------------------|-----------|-----------|---------|
| 1. (A) | 2. (D) | 3. (D) | 4. (D) | 5. (D) |
| 6. (A) | 7. (D) | 8. (A) | 9. (B) | 10. (B) |
| 11. (A) | 12. (D) | 13. (BCD) | 14. (BCD) | |
| 15. (ACD) | 16. (D) | 17. (B) | 18. (C) | 19. 8 |
| 20. $u = \sqrt{\frac{48}{5}} g\ell$ | 21. (A) q,s (B) p (C) p (D) q,r | | | |

22. (a) $K = \frac{mg}{R(3-2\sqrt{2})}$ (b) at initial instant $a_t = g$, $a_c = 0$, at bottommost position $a_t = 0$, $a_c = 0$

23. $F = -m\omega^2 r$, where r is the radius vector of the particle relative to the origin of coordinates;

$F = m\omega^2 \sqrt{x^2 + y^2}$ 24. 2 rad/sec. 25. (a) $\frac{\pi^2 R}{20 T^2} (-3\hat{i} + 4\hat{j}) \text{ m/s}^2$ (b) $\frac{\pi R}{\sqrt{2} T^2} \text{ m/s}^2$

ANSWER KEY (CAPS-7)

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|------------|-----------|----------------------------------|---------|----------|
| 1. (A) | 2. (C) | 3. (B) | 4. (B) | 5. (D) |
| 6. (B) | 7. (A) | 8. (ABC) | 9. (D) | 10. (AC) |
| 11. (ABCD) | 12. (ABC) | 13. (A) | 14. (A) | 15. (B) |
| 16. 10 | 17. a | 18. (a) 15/13 m/s (b) -9/26 m/s] | | |

19. $\frac{2v^2}{g} = 20$ 20. 25 cm 21. (A) - s ; (B) - r ; (C) - q ; (D) - p

22. (a) $v_0/3$, (b) $3\sqrt{5gR}$ 23. $\frac{\lambda\ell}{2} vg + \lambda v^3$ 24. $\eta = \frac{1}{2} \cos^2 \alpha = 0.25$

25. $m(-2\hat{i} - \hat{j} + 8\hat{k})$, $e = \frac{6}{17}$

ANSWER KEY (CAPS-8)

1. (D) 2. (D) 3. (C) 4. (B) 5. (B)
 6. (D) 7. (C) 8. (D) 9. (A) 10. (C)
 11. (A) 12. (ACD) 13. (AC) 14. (CD) 15. (B)
 16. (A) 17. (C) 18. $3w/8$ 19. 5.95 sec.
 20. (A) p,r (B) q (C) p,r,s (D) p,r,s 21. $\frac{3g}{2\sqrt{2}\ell}$ (cw) 22. $N = 2\bar{b} \sqrt{\frac{a}{b}}$

23. $t = \frac{6a\pi}{\sqrt{3}v_0}$ 24. $F = \mu_s g \left(m + \frac{7}{2}M \right)$ 25. $\therefore F_{\min} = \frac{mg \tan \alpha}{\cos \alpha + \frac{\sin^2 \alpha}{\cos \alpha}} = mg \sin \alpha$

ANSWER KEY (CAPS-9)

1. (A) 2. (D) 3. (D) 4. (A) 5. (D)
 6. (D) 7. (D) 8. (BC) 9. (BC)
 10. (ABCD) 11. (CD) 12. (D) 13. (A) 14. (A)
 15. 7 16. 4 17. 7
 18. (A) q,r,s (B) p,s (C) q,r,s (d) p,q,r,s

19. **Method - I** (a) $V_A = \sqrt{\frac{8gx}{15}(2 - \sin \theta)}$ (b) $f = \frac{mg}{15}(1 + 7 \sin \theta)$

Method - II (a) $v = \sqrt{\frac{8g(2 - \sin \theta)x}{15}}$ (b) $f = \frac{mg}{15}(1 + 7 \sin \theta)$

20. $4F/(4M + 3m)$ 21. $\omega = \frac{45}{14} = 3.21 \text{ rad/s (ccw)}, v_s = \frac{1}{7} = 0.143 \text{ m/s} \leftarrow$

22. $\frac{M}{m} = \sqrt{15}$ 23. $w_1 = F/(m_1 + 2/7m_2); w_2 = 2/7 w_1$ 24. $v = \sqrt{\frac{14gR}{3}}$

25. $2\pi \left(\frac{\alpha R^4}{4} + \frac{\beta R^5}{5} \right)$

ANSWER KEY (CAPS-10)

1. (A) 2. (B) 3. (C) 4. (B) 5. (C)
 6. (BCD) 7. (ABC) 8. (ABCD) 9. (ABD) 10. (A)
 11. (A) 12. (B) 13. (D) 14. (C) 15. (D)
 16. 5 17. 1 m 18. 1 19. 0000.50

20. (A) r,s (B) r, s (C) p (D) p,q 21. $\frac{\pi}{3} \sqrt{\frac{\ell}{g}}$ 22. $f = \frac{1}{2\pi} \sqrt{\frac{4k_1k_2}{(k_1 + k_2)m}}$

23. $A_2 = \frac{25}{46} x_0$ 24. (i) $f = \frac{1}{2\pi} \sqrt{\frac{4 \times 0.1}{0.1}} = \frac{1}{\pi} \text{ Hz}$ (ii) $V = 0.0628$ (iii) $3.9 \times 10^{-4} \text{ J}$

25. $2\pi \sqrt{\frac{\ell}{\mu g}}$

ANSWER KEY (CAPS-11)

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|---|------------------------|---|---|----------|
| 1. (A) | 2. (A) | 3. (D) | 4. (C) | 5. (A) |
| 6. (C) | 7. (C) | 8. (B) | 9. (A) | 10. (C) |
| 11. (D) | 12. (D) | 13. (D) | 14. (BC) | 15. (AC) |
| 16. (A) | 17. 3 | 18. (A) p,s (B) q,s (C) q,s (D) s | | |
| 19. $T = 2\pi \sqrt{\frac{2m\epsilon_0}{3\lambda\sigma}}$ | | 20. $V = \frac{5\rho R^2}{12\epsilon_0}$ | 21. $V = \left(\frac{9}{2} \frac{eaL}{m}\right)^{1/3}$ | |
| 22. $-\frac{8Q}{11}$ | 23. $H = \frac{4}{3}a$ | 24. $r = \frac{L}{e^{\frac{2\pi\epsilon_0 mv^2}{\lambda q}} - 1}$ | 25. $\frac{Q}{\epsilon_0} - \frac{Q}{5\epsilon_0} = \frac{4Q}{5\epsilon_0}$ | |

ANSWER KEY (CAPS-12)

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|--|--|-----------|----------|---------|
| 1. (B) | 2. (D) | 3. (A) | 4. (A) | 5. (B) |
| 6. (A) | 7. (C) | 8. (ABCD) | 9. (BCD) | |
| 10. (ABCD) | 11. (BC) | 12. (AC) | 13. (AD) | 14. (A) |
| 15. (C) | 16. 8 | 17. 2 | 18. 1 | 19. 7 |
| 20. (A) p,q,s (B) p,q,r,s, (C) p,q,s (D) p,q,r,s | | | | |
| 21. (a) 0.1 A, 4.0 V | (b) $\frac{1083.6 \times 200}{10652 \times 252} = 0.08 \text{ A}, 4.3 - \frac{1083.6}{10652} = 4.2 \text{ V}$ | | | |
| 22. $R_{AB} = \frac{\rho_p (\sqrt{7} - 1)}{3}$ | 23. (a) $R = \rho_0 \frac{L}{A} \left(\frac{e-1}{e} \right), i = \frac{V_0}{R}, (b) V(x) = \frac{V_0(e^{-x/L} - e^{-1})}{1 - e^{-1}}$ | | | |
| 24. $I_2 = \frac{20-6}{1} \text{ A} = 14 \text{ A}.$ | 25. $R_{eq} = \frac{ra}{8}(\pi + 2)$ | | | |

ANSWER KEY (CAPS-13)

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|---|---------------------------------|---|---------|---------|
| 1. (A) | 2. (C) | 3. (D) | 4. (D) | 5. (B) |
| 6. (ACD) | 7. (ABCD) | 8. (ACD) | 9. (C) | 10. (D) |
| 11. (C) | 12. (C) | 13. (D) | 14. (C) | 15. (A) |
| 16. $n = 3$ | 17. 25 | 18. 8 | 19. 300 | |
| 20. (A) – P, R ; (B) – Q ; (C) – S ; (D) – R | 21. $\frac{3}{5}$ | 22. $\beta d = 4\epsilon_0 \ell n \left(1 + \frac{\beta d}{2\epsilon_0} \right)$ | | |
| 23. $Q = \frac{1}{2} C \epsilon_2^2$. It is remarkable that the result obtained is independent of ϵ_1 . | | | | |
| 24. $i = \frac{\epsilon}{2R} \left(1 - e^{-\frac{2t}{3RC}} \right)$ | 25. $i(t) = -\frac{mg^2 t}{4E}$ | | | |

ANSWER KEY (CAPS-14)

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|---------|---------|----------|-----------|---------|
| 1. (A) | 2. (B) | 3. (B) | 4. (A) | 5. (D) |
| 6. (A) | 7. (A) | 8. (B) | 9. (B) | 10. (A) |
| 11. (D) | 12. (A) | 13. (BD) | 14. (BCD) | 15. (D) |
| 16. (B) | 17. (C) | 18. 8 | 19. 10 | 20. 2 |

21. (A) $\rightarrow$ (r) ; (B) $\rightarrow$ (q), (s) ; (C) $\rightarrow$ (s) ; (D) $\rightarrow$ (p)
22. $0, \frac{r}{2\pi} (m\omega_0^2 + 2\pi iB)$
23. $\frac{4mg}{3ia}$
24. $y = \frac{E_0 m}{qB_0^2} \left[1 - \cos \frac{qB_0}{m} t \right]$
25. $x = x_0 e^{2\pi m v / \mu_0 q l}$

ANSWER KEY (CAPS-15)

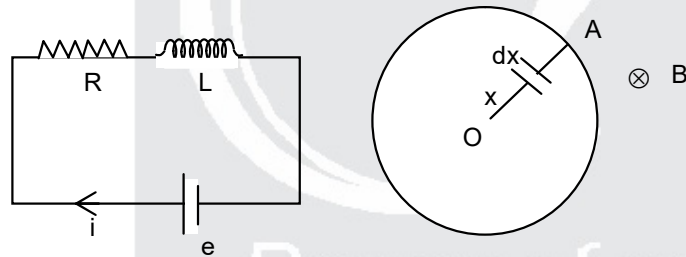
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|-----------|---|---|----------|
| 1. (C) | 2. (C) | 3. (CD) | 4. (ABC) |
| 5. (ABCD) | 6. (AC) | 7. (A) | 8. (B) |
| 9. (B) | 10. (D) | 11. (C) | 12. 67 |
| 13. 40 | 14. (A) – r,s,t ; (B) – q,r,s,t ; (C) – p,q ; (D) – q,r,s,t | 15. $P = \frac{\pi^3 B_0^2 f^2 R^4 y}{4\rho}$ | |

16. $\varepsilon = -B_0 I \beta \left(t + \frac{\alpha \times \beta^2 t^5}{4} \right)$
17. $E = \frac{1}{2} L I^2 = \frac{1}{2} \times 10 \times 10^{-3} \times (0.3)^2 \text{ J} = 4.5 \times 10^{-4} \text{ J}$

18. $\therefore \text{Peak current} = \frac{100}{9.8 \times \sqrt{2}} = 7.22 \text{ Amperes.}$

19. $v = \frac{B\ell C}{m + B^2 \ell^2 C} \left(1 - e^{-\left(\frac{B^2 \ell^2}{mR} + \frac{1}{RC} \right) t} \right)$

20. $de = B(x\omega) dx \Rightarrow e = \frac{B\omega r^2}{2}$



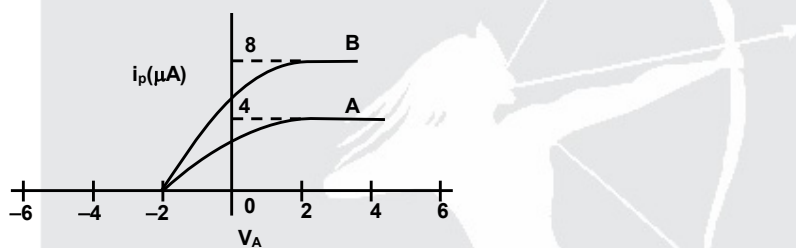
21. $\tau = \frac{mgr \cos \omega t}{2} + \frac{1}{4R} B^2 \omega r^4$
22. $F = \frac{9\mu_0^2 M^2 a^4}{4x^8 R} v$
23. $U = \frac{\pi \mu_0 r^2 i_0^2}{2R} (\because i_1 = i_2 = i_0)$

24. $U = \frac{\mu_0 i_0^2 \ell}{4\pi} \times \frac{1}{4} = \frac{\mu_0 i_0^2}{16\pi} (\because \ell = 1)$
25. $\frac{F_0}{m} x - \frac{B^2 \ell^2 a}{2km} \left\{ \ln \left(\frac{2x + \ell}{\ell} \right) \right\} + v_0 = V$

ANSWER KEY (CAPS-16)

1. (C) 2. (A) 3. (A) 4. (B) 5. (C)
6. (B) 7. (ABCD) 8. (AC) 9. (B) 10. (BC)
11. (BD) 12. (B) 13. (A) 14. (B) 15. (D)
16. 7 17. 5 18. (A) – p ; (B) – q ; (C) – s ; (D) – r.
19. (i) 4.23 Å (ii) 3.4 eV. 20. $V_1 = 0.75 \text{ V}$ 21. $1.87 \times 10^{-6} \text{ amp.}$
22. 3.376 eV.
23. Stopping potential, $V = (h\nu - W)/e = 2\text{V}$

In each case, the stopping potential is 2 V as energy of incident photon is the same. As the photon intensity is doubled, the saturation current is also doubled. The variation of i_p against V_a is sketched in the figure.



24. $F = \frac{P}{20c}$ 25. $E = \left(\frac{nh}{2\pi}\right)^6 \frac{1}{6 (Ke^2)^2 m^3}$

ANSWER KEY (CAPS-17)

1. (A) 2. (C) 3. (C) 4. (D) 5. (B)
6. (D) 7. (C) 8. (AD) 9. (B) 10. (C)
11. (A) 12. (B) 13. (A) 14. (B) 15. 2
16. 22 mg 17. 3 18. ${}_{90}^{230}\text{Th} \rightarrow {}_{88}^{226}\text{Ra} + {}_2^4\text{He} + Q ; 9.25 \text{ MeV.}$
19. $1/2.78 \times 10^6$ 20. $\text{Be}^8, E_b = 56.5 \text{ MeV.}$
21. $A_0 \left[1 - \left(\frac{5}{7}\right)^{1/4} \right]$ where $A_0 = 50 \text{ dps}$ 22. 2 23. 4
24. $\frac{Rt_{1/2}}{\ln 2} ; \frac{R}{\lambda} (1 - e^{-\lambda t})$ 25. (a) $N = \frac{1}{\lambda} [\alpha - (a - \lambda N_0) e^{-\lambda t}]$; (b) $\frac{3N_0}{2}, 2N_0]$

ANSWER KEY (CAPS-18)

1. (C) 2. (B) 3. (A) 4. (A) 5. (C)
 6. (D) 7. (A) 8. (A) 9. (ACD) 10. (ACD)
 11. (ABCD) 12. (BC) 13. (ABC) 14. (ACD) 15. (B)
 16. (B) 17. 4 18. 6 19. (A) p,q,s (B) s (C) q,r,s (D) s,t
 20. 162 vibrations/sec, 3 21. $\alpha = \frac{\rho}{Y} \frac{(v_2^2 - v_1^2)}{\Delta T}$

22. $\vec{r}_{(\text{in mm})} = 25000\hat{i} + 2\sin(100\pi t + 30^\circ)\hat{j} + 3\sin(100\pi t + 60^\circ)\hat{k}$
 $\vec{v}(\text{in m/s}) = 0.2\pi\cos(100\pi t + 30^\circ)\hat{i} + 0.3\pi\cos(100\pi t + 60^\circ)\hat{k}$
 Phase difference at time 't' is 30° constant always after they meet at 'P'.
 23. (a) $T_{\text{max}} = 1/4 m \omega^2 a_{\text{max}}^2$; (b) $T = 1/8 m \omega^2 a_{\text{max}}^2$.
 24. (i) 5 (ii) 0.2 (iii) $y_r = \frac{5}{3} A \sin$ and $y_t = \frac{2}{5} A \sin (Kx + \omega t)$.
 25. $\sin\left(\frac{\pi}{2}x + \frac{3\pi}{4} + \frac{5\pi}{2}t\right) + \sin\left(\frac{\pi}{2}x + \frac{3\pi}{4} - \frac{5\pi}{2}t\right)$

ANSWER KEY (CAPS-19)

1. (A) 2. (C) 3. (A) 4. (B) 5. (A)
 6. (B) 7. (ABCD) 8. (BD) 9. (ABD) 10. (C)
 11. (B) 12. (A) 13. (A) 14. (B) 15. (D)
 16. 336 m/s 17. 8 18. 50 19. 60
 20. $100\sqrt{15}$ m 21. $\frac{6f_0}{5}, \frac{6f_0}{7}$ 22. (A) p,q,s (B) r,s (C) s (D) r,s 23. $\theta = 60^\circ$
 24. $\langle \phi \rangle = 2\pi\ell^2 I_0 \left(1 - \frac{1}{\sqrt{1 + (R/\ell)^2}}\right) = 20 \mu\text{W}$.
 25. (a) 1/5 m, 50 Hz, 10 m/s; (b) 100 Pa, 10π Pa, (c) 1/20 m

ANSWER KEY (CAPS-20)

1. (C) 2. (B) 3. (B) 4. (D) 5. (C)
 6. (D) 7. (ABCD) 8. (AB) 9. (ABCD) 10. (ABC)
 11. (ABC) 12. (AC) 13. (ABD) 14. (C) 15. (B)
 16. (C) 17. 21 18. $v = 4 \text{ m/s}$ 19. 14
 20. (A) p,r,s (B) q (C) p,r,s (D) q,r 21. $\frac{2q}{5(mg + P_0 A)}$ 22. $W = p_0 V_0 \ln [(\eta + 1)^2 / 4\eta]$
 23. $K = 2000 \text{ N/m}$, $Q = 1297 \text{ Joules approx.}$
 24. $P = \frac{10}{3} \times 10^5 \text{ N/m}^2$, $T = \frac{10500}{31} \text{ K} \approx 338.71 \text{ K}$ 25. $\frac{V_0}{2^{\frac{\gamma+1}{\gamma}}}$ in each cases

ANSWER KEY (CAPS-21)

1. (B) 2. (C) 3. (A) 4. (C) 5. (ACD)
 6. (BC) 7. (A) 8. (C) 9. (B) 10. (B)
 11. (A) 12. (C) 13. 273 K. 14. 4 15. 7

16. $\ell_0 \left[1 + \frac{2Y_a \alpha_a + Y_s \alpha_s}{2Y_a + Y_s} \theta \right]$

17. $F = \frac{AT(L_1 \alpha_1 + L_2 \alpha_2)Y_1 Y_2}{L_1 Y_2 + L_2 Y_1}$, Length of the first rod = $L_1 + \frac{L_1 L_2 T(Y_1 \alpha_1 - Y_2 \alpha_2)}{L_1 Y_2 + L_2 Y_1}$,

Length of the second rod = $L_2 + \frac{L_1 L_2 T(Y_2 \alpha_2 - Y_1 \alpha_1)}{L_1 Y_2 + L_2 Y_1}$

18. 10.368 s 19. $\frac{3K_0 A(\theta_2 - \theta_1)}{8l}$ 20. $\frac{K \pi R_1 R_2 (T_H - T_C)}{L}$
 21. $\Delta T = (\Delta T)_0 e^{-\alpha t}$, where $\alpha = (1/C_1 + 1/C_2) SK/l$ 22. 270.00, 120.00
 23. (A) – (p) ; (B) – (r) ; (C) – (s) ; (D) – (q)

ANSWER KEY (CAPS-22)

1. (C) 2. (C) 3. (B) 4. (D) 5. (D)
 6. (B) 7. (B) 8. (D) 9. (C) 10. (ABC)
 11. (B) 12. (B) 13. (C) 14. 30 15. 12
 16. 1 17. 520 nm 18. 1100 19. 2311
 20. (A) P, R, S (B) P, Q, S (C) P, Q, S (D) P, R, S
 21. (a) 80 cm behind the lens (b) 4 mm (c) $\beta = 60 \mu\text{m}$ 22. $\lambda = 600 \text{ nm}$, $t = 24 \mu\text{m}$
 23. (a) $\Delta \phi = \left(\frac{1}{\ell} + \frac{\mu}{D} \right) \frac{\pi d^2}{\lambda}$ (b) $\Delta \phi = \left(\frac{\mu}{\ell} + \frac{1}{D} \right) \frac{\pi d^2}{\lambda}$; $D_{\min} = \frac{\beta}{2} = \frac{\lambda D}{2d}$ 24. 0, 1.5 mm 25. I

ANSWER KEY (CAPS-23)

1. (A) 2. (D) 3. (A) 4. (B) 5. (A)
 6. (D) 7. (A) 8. (B) 9. 3 10. 3
 11. 4 12. (A) – p ; (B) – q ; (C) – p ; (D) – p 13. $\frac{g A \rho}{4} (h_1 - h_2)^2$
 14. $v = \sqrt{\frac{2(\rho_m - \rho_a)gh}{\rho_a}}$ 15. $\frac{7m}{3\lambda}$ 16. $\frac{4T}{\rho v^2}$ 17. $Q = \sqrt{768\pi^2 R^3 \epsilon_0 T}$
 18. $\theta = 45^\circ$ 19. $t = 1 \text{ sec}$ 20. $T = \left(\frac{P_0 h}{\ell - h} + \rho g h \right) \frac{d}{4 \cos \theta}$

ANSWER KEY (CAPS-24)

1. (B) 2. (AD) 3. $\frac{\eta A v_0}{D}$ 4. 4 mm 5. (B)
6. $t = 100 \ln 2$ seconds 7. $t = \frac{m(R_2 - R_1) \ln 2}{4 \pi \eta \ell R_1}$ 8. 2
9. 32 10. 3 11. $\Delta R = \frac{k \lambda Q_0}{A Y}$ 12. $\eta = 0.77 \text{ N-sec/m}^2$.
13. 1.33 cm 14. $\frac{1000}{9} \text{ N/mm}^2, 300 \text{ mm}^2, \frac{1000}{3} \text{ mm}^2$ 15. 9mm

16. $d = \int_0^\ell \frac{F x dx}{A \ell Y} = \frac{F \ell}{2 A Y}$

ANSWER KEY (CAPS-25)

1. (B) 2. (A) 3. (A) 4. (A) 5. (A)
6. (A) 7. (AB) 8. (AB) 9. (AC) 10. (AD)
11. (AC) 12. (C) 13. (A) 14. 3 15. 7
16. 8 17. $E = T + U = -GmM_s/2a$, where m_s is the mass of the Sun.
18. $\frac{dp}{dr} = -\frac{GM_r}{r^2} \rho_r$

